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Reading the Score, Not the Shelf

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Abstract. Handheld barcode scanners increasingly assign each pick a live confidence score, using a picker's rolling average to trigger automatic retraining once it falls below a set threshold. Whether the score tracks a picker's actual accuracy, independent of the system that generates it, has not previously been examined. We link scanner-logged confidence scores across 64 pickers at nine distribution centres to a concurrent programme of hand-conducted inventory audits, blind to the scanner's own readings, covering 11,842 audited picks over five months. Confidence score shows a negligible relationship with audit-verified pick accuracy ($r = 0.06$, n.s.) but a strong, near-linear relationship with the number of weeks since the picker's most recent retraining session ($r = 0.81$), a relationship that holds regardless of whether that retraining changed the picker's error rate at all. Error rates for mis-picks, wrong quantities, and wrong-bin placements are statistically indistinguishable in the fortnight before and after a retraining event. We report the audit-score divergence as the null result it is, and discuss what a rolling confidence estimator keeps counting once its retraining calendar has quietly become its dominant input.

Introduction

Handheld scanning platforms across distribution and fulfilment operations increasingly report more than a barcode read: each scan carries a live confidence score, typically built from read clarity, scan latency, and a picker's short-term error history, updated as a rolling average. When that average falls below an operator-set threshold, the platform flags the picker for a retraining module, and the score itself is frequently reported upward as a proxy for pick accuracy on operational dashboards. The appeal is straightforward: a number already being computed for one purpose becomes available for another, and an institution that already trusts the platform for scanning has reason to extend that trust to the platform's own account of how well the scanning is going.

That extension has, to our knowledge, not been checked. A confidence score computed by the same system it purports to monitor is not obviously equivalent to an accuracy measure computed independently of that system, and the two have rarely if ever been placed side by side under audit conditions blind to the scanner's own logs. We report what happens when they are.

We link scanner-logged pick-confidence scores across 64 pickers at nine distribution centres in a single logistics network to a concurrent programme of manual inventory audits, conducted by staff with no access to the scanner's own confidence readings, over five months and 11,842 audited picks. Our contributions:

- We assemble what we believe is the first blind, independently audited comparison of scanner-reported pick confidence against ground-truth error rates across multiple sites.
- We show that the score's principal predictor is administrative rather than behavioural: time since the picker's last retraining session, not any change in what the picker actually does.
- We discuss the consequences for institutions that report scanner confidence upward as an accuracy metric, and for the retraining programmes the score is used to trigger.

Related work

Barcode and RFID scanning are established correctives to manual data-entry error in warehouse and retail inventory, and the accuracy gains are well documented at the record level [1], [2]. Less studied is the confidence metadata a scanning platform now typically attaches to each read; where scanning literature asks whether a scan happened correctly, we ask what a platform's own running estimate of correctness tracks once the estimate leaves the scan and starts governing something else.

Human-factors work on manual order picking documents a range of failure modes — physical and mental fatigue, memory demand, communication and supervision gaps — that a purely device-side confidence signal cannot observe [3]. A separate literature on algorithmic management describes how logistics operators route decisions about training, discipline, and pace through platform-generated metrics, with workers adapting their visible behaviour to what the metric can see rather than to the underlying task [4], [5].

The automation literature offers two relevant threads. Trust-calibration work asks whether a worker's confidence in a system tracks the system's actual reliability [6], [7], [8]; we ask the mirror question, whether the system's own confidence in the worker tracks the worker's actual reliability. Vigilance research documents how human monitoring performance degrades under automation, particularly once the task grows monotonous enough to underload the person still nominally watching it [9]. Skill-decay research establishes that retraining effects on true performance follow their own, generally slow, curve [10], [11], [12] — a curve our data suggest the confidence score does not track. Where an institutional indicator drifts from the process it was built to represent, longstanding measurement theory has a name for the mechanism [13], [14], and a caution about mistaking a change in what is watched for a change in what is done [15].

Two prior studies within the University's own corpus bear on the underlying mechanism from opposite sides of the screen. Displaying a boom gate's automatic-approval confidence score to security staff has been shown to suppress their independent scrutiny precisely in the band where the score reads highest [16]; separately, lifeguards' own independent scanning of the water has been shown to decline over tenure under an installed automated drowning-detection alarm, even as the alarm's own compliance reporting holds at ceiling throughout [17]. Both concern a human adjusting to a machine's stated confidence; the present study asks the question the other way, treating the machine's confidence in the human as the thing to be audited.

Method

Data were collected across nine distribution centres in a single logistics network operating a common handheld scanning platform, covering 64 pickers whose tenure at the time of enrolment ranged from three weeks to six years. The platform computes a per-scan confidence value C_t from three components — c_t (barcode read clarity), h_t (a hesitation penalty derived from time-to-scan), and s_t (a deviation penalty against the picker's own baseline scan speed) — combined as

$$C_t = 100(w_1 c_t + w_2(1 - h_t) + w_3(1 - s_t))$$

and reported to supervisors as a rolling exponential average

$$\hat{C}_t = \lambda C_t + (1 - \lambda)\hat{C}_{t-1}.$$

When $\hat{C}_t$ falls below a site-configured threshold (68 across all nine sites during the study window), the platform automatically enrols the picker in a half-day retraining module and resets the display, though not the underlying scan history.

Independent of the platform, each site's inventory team ran a stratified cycle-count audit, sampling pick locations at random within velocity strata and physically verifying the item, quantity, and bin against the order record. Auditors worked from a printed pick list stripped of any confidence annotation and had no access to the scanning platform's dashboard; a separate reconciliation step, run after each audit cycle closed, matched audited picks to their originating scanner session via an anonymised picker ID that carried no confidence data. Over five months this yielded 11,842 audited picks, distributed roughly evenly across the nine sites and across the range of $\hat{C}_t$ values observed in the platform's own logs.

For each audited pick we recorded the scanner's contemporaneous $\hat{C}_t$ reading, the audit outcome (correct, mis-pick, wrong quantity, or wrong bin), and, separately, the number of completed weeks since the picker's most recent retraining event, drawn from the platform's own enrolment log. Error type followed the platform's existing taxonomy; audits used dual sign-off on any discrepancy to guard against auditor error, with a 4% re-audit sample confirming inter-auditor agreement above 92%.

Results

Confidence score showed a negligible relationship with audit-verified accuracy across the full sample (Pearson $r = 0.06$, $p = .31$; Figure 1). Binning picks into eleven confidence-score bands and computing each band's audit-verified error rate produces a line indistinguishable from flat across the platform's full operating range: the highest-scoring band's error rate (3.8%) sits inside the lowest-scoring band's confidence interval (4.6%), and no band differs from the sample mean by more than a single point.

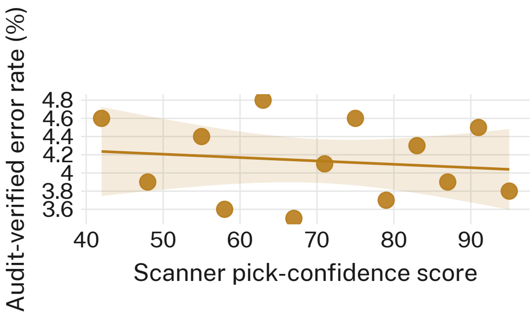

Figure 1: Audit-verified error rate against scanner confidence score, by score band ($n = 11,842$ picks); the fitted line's slope is not distinguishable from zero.

The score does track something, only not that. Mean confidence rises steadily for the first six weeks following a retraining event — from 54 at week 0 to 86 by week 6 — before plateauing around 88 for the remainder of the observed window (Figure 2). The relationship between $\hat{C}_t$ and weeks-since-retraining is strong (Pearson $r = 0.81$, $p < .001$) and closely tracks the shape of the rolling-average update rule itself: with λ fixed across all nine sites, six weeks is approximately the interval over which $\hat{C}_t$ converges from a post-retraining reset toward its ceiling, independent of any single picker's day-to-day accuracy.

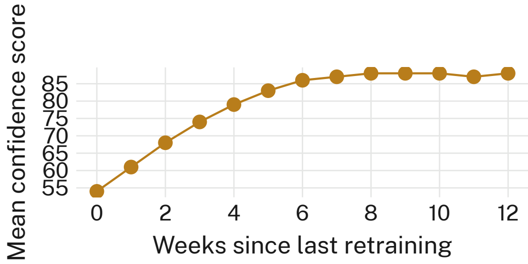

Figure 2: Mean scanner confidence score by weeks since the picker's last retraining session ($n = 64$ pickers); the curve's shape matches the rolling-average window, not an independently observed accuracy trend.

Retraining itself changed the score without changing the thing the score is meant to summarise. Comparing audit-verified error rates in the two-week windows immediately before and after a retraining event, mis-pick rate moved from 2.1% to 2.3%, wrong-quantity error from 1.4% to 1.2%, and wrong-bin placement from 0.6% to 0.7% — no comparison approached significance (all $p > .4$; Figure 3), against a confidence-score jump across the same fortnight that averaged 19 points.

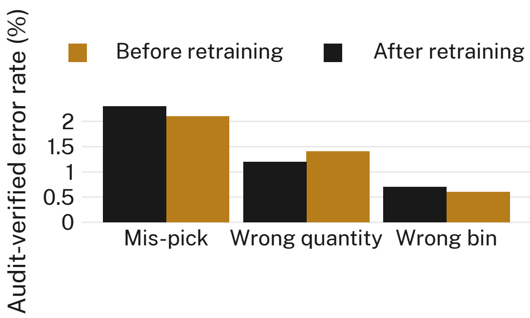

Figure 3: Audit-verified error rate by error type, the fortnight before versus after a retraining event ($n = 1,206$ retraining events); none of the three comparisons is statistically distinguishable.

A supplementary model regressing audit-verified error rate on confidence score, controlling for site, tenure, and error type, added no explanatory power over tenure alone ($\Delta R^2 = .004$); a matching model with weeks-since-retraining in place of confidence score performed identically to the confidence score itself, consistent with the two being close to interchangeable as predictors of one another rather than of the audited outcome.

Discussion

Taken together, these results are consistent with a confidence score whose dominant input has become the platform's own retraining calendar rather than any independently observable change in picker accuracy. This is not a claim that the underlying per-scan signal — read clarity, hesitation, speed deviation — is uninformative; rather, once aggregated into a rolling average anchored to a reset event, the aggregate inherits the reset's clock more than the picker's behaviour. Institutions that report $\hat{C}_t$ upward as a pick-accuracy indicator are, on this evidence, substantially reporting time-since-retraining under an accuracy label, a substitution the platform's own dashboard gives no way to distinguish.

The retraining programme itself is not without value on other grounds — procedural refreshers plausibly support outcomes this audit did not measure, including picker confidence, onboarding speed for new stock lines, or compliance with updated safety procedure — but its effect on the specific outcome the confidence score is built to flag was not detectable in this sample. Where a score generated by a system is used to certify that same system's users without an external check, our results suggest the certification and the system's own bookkeeping may be difficult to tell apart from the outside, and from a supervisor's dashboard, impossible to tell apart at all.

Limitations

Our audit sample is drawn from nine distribution centres within a single logistics network, all running the same scanning platform and rolling-average configuration; a platform using a different aggregation window might show a different post-retraining curve, though the consistency of the six-week convergence across nine independently managed sites suggests the pattern reflects the update rule rather than any one site's practice. Auditors, though blind to the platform's confidence readings, were not blind to which pickers had recently been retrained, which their protocol could not conceal given rostering; however, an audit outcome is a physical count against a pick list, not a judgement call, which limits the scope for that awareness to shape the result reported here.

Conclusion

We audited a widely deployed pattern — a handheld scanner's rolling confidence score, used both to trigger retraining and to report pick accuracy upward — against inventory counts collected independently of the scanning platform, and found the score tracks the retraining calendar far more closely than it tracks the accuracy it is presented as measuring. The finding is a null result on the question that matters operationally: retraining did not measurably change error rates in this sample. Future work might test whether the same decoupling appears in platforms using a different rolling-average window or reset condition, and whether supervisors' operational decisions change once the audit-score gap reported here is made visible to them rather than left in a research paper.

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